



GPU vs CPU

than the CPU. Though it might give an impression that running 40 such CPUs would get you the power of that single GPU, it would be wrong. Reasons:

1. GPUs come with dedicated hardware logic to calculate upon 3D data which makes them even more powerful than the CPU when it comes to computing 3D objects.
2. GPUs would get snail paced if you asked them to do what the CPU does. This is because GPUs are Single Instruction, Multiple Data (SIMD) systems. What this means is that GPUs will show their true monstrous number crunching abilities only when one single instruction can be executed on an array of similar data. Have a look at the inset note about SIMD to learn more.
3. GPUs would be slow if they were to use the normal methods used to access data in RAM. The amount of computation they do requires excessively fast memory I/O and their memory architecture is built to serve just that. More about this shortly.

SIMD systems, though come with multiple processors, they require that all processors be working on the same instruction at a time, however on multiple data. Hence, if one processor in a SIMD system is adding two numbers, other processors would also be adding two numbers, however a different set of numbers. SIMD offers excessive performance gains at the cost of programmer having to take care of intrinsic details of how the algorithm would run.



GTX 580 desktop PCB

The graphics memory

As we just said, graphics cards are able to compute at such speeds largely because of how they access the graphics memory. The CPU RAM module is accessed serially i.e. the RAM modules of the